Chapter 8 — Physical Optics and Gravitational Waves

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Polarization proves light is: (a) longitudinal (b) stationary (c) transverse (d) mechanical
2. Malus's law: (a) $I = I_0 \tan^2\theta$ (b) $I = I_0 \cos\theta$ (c) $I = I_0 \sin^2\theta$ (d) $I = I_0 \cos^2\theta$
3. Brewster's law: (a) $\cos\theta_p = n$ (b) $\tan\theta_p = 1/n$ (c) $\sin\theta_p = n$ (d) $\tan\theta_p = n$
4. Crossed polaroids transmit: (a) no light (b) half light (c) double light (d) maximum light
5. Gravitational waves travel at: (a) c/2 (b) zero (c) sound speed (d) c
6. First direct GW detection: (a) 2017 (b) 2012 (c) 2015 (d) 1916
7. LIGO uses: (a) laser interferometry (b) radio telescope (c) lens camera (d) sonar
8. LIGO arm length: (a) 10 km (b) 1 km (c) 3 km (d) 4 km

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why can longitudinal waves not be polarized?
2. Why does Malus's law contain cos squared rather than cos?
3. Why are polaroids used in sunglasses?
4. Why can we not detect gravitational waves from everyday objects?
5. Why are very long arms used in LIGO?
6. Why are two detectors such as LIGO and Virgo used together?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Explain how plane-polarized light is produced and detected by polaroids. **(4)**
(b) State Brewster's law and Malus's law and explain their use. **(4)**
Q.4 (a) Describe gravitational waves and their four basic types. **(4)**
(b) Explain how a laser interferometer detects gravitational waves. **(4)**

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